Adithya Rajendran

Field Software Engineer

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PROFESSIONAL SUMMARY

Field Engineer with two years' experience designing, deploying, and maintaining secure private clouds at scale. Adept at architecting and leading large-scale OpenStack, Kubernetes, and Ceph deployments for enterprise clients. Highly experienced with automation using Python and Terraform. Known for delivering custom, client-focused solutions through cross-team collaboration, technical troubleshooting, and a strong focus on process optimization and knowledge sharing.

SKILLS

- **Technical Skills**: OpenStack, Kubernetes, Ceph, Linux/Unix, Virtualization (VMware, KVM, Xen), Docker, Active Directory, OIDC, Networking, AWS (EC2, S3, RDS, Lambda), Git, Terraform, PXE, IPMI, DHCP
- **Security Skills**: Wazuh (SEIM), Okta Auth0 (IAM), MITRE ATT&CK, NIST SP-800, OPNsense (Firewalls), Snort (Intrusion Detection System), Nmap, Compliance (CIS, FIPS, DISA-STIG), Privilege Escalation, Penetration Testing
- **Software Development Skills**: C/C++, Python, Go, HTML, CSS, RESTful APIs, SQL (PostgreSQL, MySQL), NoSQL

WORK EXPERIENCE

Canonical (Ubuntu) **May 2024 – Present**
Field Software Engineer I Fremont, CA

- Architected and led the deployment of large-scale private cloud solutions (OpenStack, Kubernetes, Ceph) for over a dozen enterprise clients, reducing the average Total Cost of Ownership (TCO) by 70%.
    - Created automations using Python, Terraform, and Bash, reducing errors and manual effort.
    - Partnered with core engineering and support teams to architect custom solutions for unique customer scenarios, enhancing system performance and reducing support escalations.
    - Authored custom CIS and DISA-STIG tailoring files to assist in meeting compliance requirements like CMMC and HIPAA, as well as strengthen the overall security posture.
- Troubleshot issues across the entire suite, from the kernel to the networking and switches, all the way to applications running on the OS, providing patches to bugs where necessary.
    - Documented root cause analyses and solutions in the knowledge base, improving time-to-resolution.
- Partnered with the sales team in the pre-sales process, leading solution-oriented discovery calls, delivering customized demonstrations, and advising clients on alternative strategies to better address their business goals.
    - Assisted with drafting Statement of Work documents while ensuring alignment between sales, support, and engineering teams, effectively providing a solid outline of the engagement.
- Collaborated with product management and engineering teams by providing critical field feedback on customer use-cases and technical challenges, influencing the product roadmap.
- Mentored new field engineers on technical troubleshooting and customer engagement strategies.

Technical Consulting & Research, Inc. (TCR) **December 2023 – May 2024**
Cybersecurity Analyst Intern Fremont, CA

- Addressed 4+ cloud provider compliance issues via fit-gap analysis, improving regulatory adherence.
- Leveraged the MITRE ATT&CK framework to refine threat modeling processes, achieving a notable 15% reduction in system vulnerabilities and bolstering security defenses.
    - Applied security best practices such as the principle of least privilege and network segmentation
- Developed automation scripts in Python, reducing the need to direct human work hours by 1-2 hrs. per week.
- Built demo applications in JavaScript showing the effects and operation of compliance requirements.
- Orchestrated secure authentication and authorization with Okta's Auth0, enhancing user access control.
- Delivered detailed security reports and visualizations, significantly raising non-technical stakeholder awareness.
- Presented at the IGNITE and Bucknell University conferences on "Navigating AI Risks for Small Businesses."

PROJECTS

Home-Lab with Segmented Networks and High Availability

- Engineered a robust personal cloud environment to master and apply core principles of modern IT infrastructure, from bare-metal server provisioning to secure application deployment.
- Segmented network traffic with VLANs and enforced access control with policy-based firewall rules.
- Optimized for high availability and performance by configuring link aggregation for network redundancy and jumbo frames for efficient Ceph replication.

Kubernetes Cluster with NFS and OIDC

- Built a highly available, three-node Kubernetes cluster designed for resilient app deployment and management, resulting in zero downtime during upgrades and maintenance.
- Proactively secured the environment by applying CIS Level 1 hardening profiles.
- Integrated Okta (OIDC) for authentication and NFS for persistent storage, creating a production-ready platform.

Gmail Spam Filter using AI for Detection

- Developed a custom application that integrates directly with the Gmail API to automate the process of fetching and analyzing incoming emails, achieving a 90% spam detection accuracy rate.
- Engineered the detection model using a Multi-Layer Perceptron (MLP) classifier and fine-tuned the architecture, which led to reduced false positives and faster email classification.
- This project showcases practical skills in machine learning, API integration, and building automated solutions for real-world data challenges.
- GitHub: https://github.com/Adithya-Rajendran/Gmail-Filter

Applied Penetration Testing & Ethical Hacking

- Achieved a global ranking within the top 3% on TryHackMe and the top 700 on HackTheBox, platforms used by industry professionals for continuous skill validation.
- Demonstrated practical expertise by successfully navigating real-world security scenarios, mastering techniques in vulnerability assessment, network exploitation, and privilege escalation.

NextJS Portfolio Website + Blogs

- Developed a personal portfolio website utilizing Next.js and React, showcasing my software development projects and technical blogs.
- Implemented server-side rendering for SEO optimization and faster page loads, enhancing user experience.
- GitHub: https://github.com/Adithya-Rajendran/portfolio-website

Rust-based DNS server with intelligent Caching (In-Progress)

- Developing a DNS resolver from scratch in Rust, leveraging the language's strengths in performance and memory safety to build a highly reliable service.
- Designing and implementing an intelligent caching mechanism to minimize latency and reduce upstream queries, significantly improving resolution speed for frequently accessed domains.
- Engineering security at the network edge by integrating traffic filtering capabilities and forwarding DNS logs to an external SIEM for enhanced threat detection and network observability.

CERTIFICATIONS

- Offensive Security Certified Professional — Currently Preparing
- AWS Certified Solutions Architect — September 2023 - September 2026
- CompTIA Security+ — August 2022 - August 2025

EDUCATION

University of California, Santa Cruz — **Graduated 2023**

Bachelor of Science in Computer Science — Santa Cruz, CA

- **Relevant Coursework**: Computer Systems and C, Cryptography, Advanced Computer Networking, Principle of System Design, Artificial Intelligence, Natural Language Processing, Data Structures and Algorithms